Antoine Vuignier

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Final-year PhD candidate in theoretical physics, transitioning to applied research in machine learning. My work focuses on turning abstract questions into well-defined mathematical problems and producing quantitative answers, using a combination of analytical reasoning and computational methods. I'm currently exploring research-oriented roles where strong quantitative thinking and technical problem-solving are central.

Education

2022 – ... ◇ **PhD, EPFL - Fields and Strings Laboratory**
Supervised by João Penedones. Research on the emergence of spacetime in quantum gravity and non perturbative approaches to quantum field theory. Built computational workflows combining Mathematica and Python for symbolic algebra, numerical linear algebra, and large-scale convex optimization (semidefinite programming) on external clusters.
Thesis: *What is spacetime made of ?*

2017 – 2022 ◇ **Bachelor and Master of Physics, EPFL**

Skills

General ◇ Problem solving, critical thinking, public speaking, teaching, project supervision

Languages ◇ Native in French, fluent in English, B2 in German.

Coding ◇ Python (NumPy, SciPy, JAX, matplotlib),C++, Mathematica, LaTeX. Experience with numerical optimization, symbolic computation and cluster based workflows.

Research Output

2022-2025 ◇ 5 publications in high energy theoretical physics (JHEP), see INSPIRE profile `https://inspirehep.net/authors/2612961`.

2024-2025 ◇ Multiple invited talks at international workshops and universities

Workshops and Training

2025 ◇ **Artificial Intelligence for High Energy Physics** - Workshop Hands-on training with modern ML frameworks (JAX). Studied neural network architectures, optimization methods, and their applications to physics problems.

2022-2025 ◇ Attended 10+ international schools and workshops in theoretical physics and mathematics.

Teaching

2019-2025 ◇ **Teaching Assistant, EPFL** - Courses: general physics and mathematics (bachelor level), classical electrodynamics, statistical physics, quantum physics, relativity and cosmology. Also taught at the EPFL summer school on general relativity for high school students.

Scientific Activities

2025 ◇ Organized weekly seminars at EPFL.

2023-2024 ◇ Supervised two master students (M. Nabavi and B. Tafreshi) on their master projects at EPFL.

2022-2025 ◇ Gave two outreach talks making my research accessible to high school and first year university students